
MI 2020 PU3 H2 MATHS 9758/01 PRELIM EXAM SOLUTION

Qn	Solution
1	$\ln(x^2 - 4) + x^2 - 8 \leq 0$ $\ln(x^2 - 4) \leq 8 - x^2$ The suitable additional curve is $y = 8 - x^2$. Intersection points between the 2 graphs: $(-2.63, 1.07)$, $(2.63, 1.07)$ (3 s.f.) The solution to the inequality $\ln(x^2 - 4) + x^2 - 8 \leq 0$: $-2.63 \leq x < -2$ or $2 < x \leq 2.63$ (3 s.f.)
2 (i)	Let A be the surface area of a spherical bubble of radius r. Then $A = 4\pi r^2$. Working 1: Without making r the subject of the formula Method 1: Implicit differentiation $\frac{dA}{dt} = 4\pi \left(2r \frac{dr}{dt} \right) = \pi \Rightarrow \frac{dr}{dt} = \frac{1}{8r}.$ Method 2: Direct differentiation and chain rule $\frac{dA}{dr} = 8\pi r.$ $\frac{dr}{dt} = \frac{dr}{dA} \frac{dA}{dt} = \frac{1}{8\pi r} (\pi) = \frac{1}{8r}.$

Qn	Solution
	Working 2: Make r the subject of the formula $A = 4\pi r^2 \Rightarrow r = \sqrt{\frac{A}{4\pi}}. \text{ (reject } r = -\sqrt{\frac{A}{4\pi}} \text{ since } r > 0)$ $r = \left(\frac{A}{4\pi}\right)^{\frac{1}{2}} \text{ or } \frac{A^{\frac{1}{2}}}{2\sqrt{\pi}}.$ Method 3: Implicit differentiation $\frac{dr}{dt} = \frac{1}{2} \left(\frac{A}{4\pi}\right)^{-\frac{1}{2}} \frac{1}{4\pi} \frac{dA}{dt} \text{ or } \frac{1}{2\sqrt{\pi}} \frac{1}{2} A^{-\frac{1}{2}} \frac{dA}{dt}$ $\frac{dr}{dt} = \frac{\pi}{4\sqrt{\pi A}} \text{ (since } \frac{dA}{dt} = \pi)$ $\frac{dr}{dt} = \frac{\pi}{4\sqrt{\pi(4\pi r^2)}} = \frac{1}{8r}.$ Method 4: Direct differentiation and chain rule $r = \left(\frac{A}{4\pi}\right)^{\frac{1}{2}} \text{ or } \frac{A^{\frac{1}{2}}}{2\sqrt{\pi}}$ $\frac{dr}{dA} = \frac{1}{2} \left(\frac{A}{4\pi}\right)^{-\frac{1}{2}} \frac{1}{4\pi} \text{ or } \frac{1}{2\sqrt{\pi}} \frac{1}{2} A^{-\frac{1}{2}}$ $\frac{dr}{dt} = \frac{dr}{dA} \frac{dA}{dt} = \frac{\pi}{4\sqrt{\pi A}} \text{ (since } \frac{dA}{dt} = \pi)$ $\frac{dr}{dt} = \frac{\pi}{4\sqrt{\pi(4\pi r^2)}} = \frac{1}{8r}.$
2 (ii)	Let V be the volume of a spherical balloon of radius r. Then $V = \frac{4}{3}\pi r^3$. Method 1: Implicit differentiation $\frac{dV}{dt} = \frac{4}{3}\pi \left(3r^2 \frac{dr}{dt}\right) = 4\pi r^2 \left(\frac{1}{8r}\right) = \frac{1}{2}\pi r.$ Method 2: Direct differentiation and chain rule $\frac{dV}{dr} = \frac{4}{3}\pi (3r^2) = 4\pi r^2.$ $\frac{dV}{dt} = \frac{dV}{dr} \frac{dr}{dt} = 4\pi r^2 \left(\frac{1}{8r}\right) = \frac{1}{2}\pi r.$ When $V = \frac{288}{\pi^2} \Rightarrow \frac{4}{3}\pi r^3 = \frac{288}{\pi^2} \Rightarrow r^3 = \frac{216}{\pi^3} \Rightarrow r = \frac{6}{\pi}$. Hence, $\frac{dV}{dt} = \frac{1}{2}\pi \left(\frac{6}{\pi}\right) = 3$. Required rate of change = $3 \text{ cm}^3/\text{s}$.

Qn	Solution
3 (i)	Method 1: $u_r - u_{r+1} = \frac{1}{(r+1)!} - \frac{1}{(r+2)!}$ $= \frac{r+2-1}{(r+2)!}$ $= \frac{r+1}{(r+2)!} \text{ (shown)}$ Method 2: $u_r - u_{r+1} = \frac{1}{(r+1)!} - \frac{1}{(r+2)!}$ $= \frac{(r+2)! - (r+1)!}{(r+1)!(r+2)!}$ $= \frac{(r+2)(r+1)! - (r+1)!}{(r+1)!(r+2)!}$ $= \frac{(r+1)![r+2-1]}{(r+1)!(r+2)!}$ $= \frac{r+1}{(r+2)!} \text{ (shown)}$
3 (ii)	Method 1: $\sum_{r=1}^n \frac{r+1}{(r+2)!} = \sum_{r=1}^n [u_r - u_{r+1}]$ $= \sum_{r=1}^n \left[\frac{1}{(r+1)!} - \frac{1}{(r+2)!} \right]$ $= \left[\frac{1}{2!} - \frac{1}{3!} \right]$ $+ \frac{1}{3!} - \frac{1}{4!}$ $+ \frac{1}{4!} - \frac{1}{5!}$ $\dots$ $+ \frac{1}{(n)!} - \frac{1}{(n+1)!}$ $+ \frac{1}{(n+1)!} - \frac{1}{(n+2)!} \Big]$ $= \frac{1}{2!} - \frac{1}{(n+2)!}$ $= \frac{1}{2} - \frac{1}{(n+2)!}$

Qn	Solution								
	Method 2: $\sum_{r=1}^n \frac{r+1}{(r+2)!} = \sum_{r=1}^n [u_r - u_{r+1}]$ $= [u_1 - u_2 + u_2 - u_3 + u_3 - u_4 + \dots + u_{n-1} - u_n + u_n - u_{n+1}]$ $= u_1 - u_{n+1}$ $= \frac{1}{2!} - \frac{1}{(n+2)!}$ $= \frac{1}{2} - \frac{1}{(n+2)!}$								
3 (iii)	$S_{\infty} = \sum_{r=1}^{\infty} \frac{r+1}{(r+2)!} = \frac{1}{2}$ [Reason: As $n \rightarrow \infty$, $\frac{1}{(n+2)!} \rightarrow 0$, $S_n = \frac{1}{2} - \frac{1}{(n+2)!} \rightarrow \frac{1}{2}$] Method 1 $ S_n - S_{\infty} \leq 10^{-4}$ $\left \frac{1}{2} - \frac{1}{(n+2)!} - \frac{1}{2} \right \leq 0.0001$ $\left \frac{1}{(n+2)!} \right \leq 0.0001$ <table border="1" data-bbox="327 1429 802 1671"> <tr> <th>n</th><th>$\left \frac{1}{(n+2)!} \right$</th></tr> <tr> <td>5</td><td>1.98×10^{-4}</td></tr> <tr> <td>6</td><td>2.48×10^{-5}</td></tr> <tr> <td>7</td><td>2.76×10^{-5}</td></tr> </table> The smallest value of n is 6. Method 2 $S_{\infty} - 10^{-4} \leq S_n \leq S_{\infty} + 10^{-4}$ $\frac{1}{2} - 10^{-4} \leq \frac{1}{2} - \frac{1}{(n+2)!} \leq \frac{1}{2} + 10^{-4}$ $0.4999 \leq \frac{1}{2} - \frac{1}{(n+2)!} \leq 0.5001$	n	$\left \frac{1}{(n+2)!} \right $	5	1.98×10^{-4}	6	2.48×10^{-5}	7	2.76×10^{-5}
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4 (i)	Method 1: Subtraction of 2 areas Required area = $\int_0^1 \left(\frac{10}{2+3x^2} \right) - (4x-2) dx$ ≈ 3.61739 $= 3.617 \text{ units}^2 \text{ (3 d.p.)}$ Method 2: Addition of 2 areas For $y = \frac{10}{2+3x^2}$, when $x = 0$, $y = 5$. For $y = 4x - 2$, when $x = 0$, $y = -2$. $y = 4x - 2 \Rightarrow x = \frac{y+2}{4}$ $y = \frac{10}{2+3x^2} \Rightarrow x = \sqrt{\frac{1}{3} \left(\frac{10}{y} - 2 \right)} = \sqrt{\frac{2}{3} \left(\frac{5}{y} - 1 \right)}$ Required area = $\int_{-2}^2 \frac{y+2}{4} dy + \int_2^5 \sqrt{\frac{2}{3} \left(\frac{5}{y} - 1 \right)} dy$ ≈ 3.61739 $= 3.617 \text{ units}^2 \text{ (3 d.p.)}$ Note: $\int_{-2}^2 \frac{y+2}{4} dy = \text{Area of triangle} = \frac{1}{2}(1)[2 - (-2)] = 2$																

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4 (ii)	For $y = \frac{10}{2+3x^2}$, when $x = 0$, $y = 5$. For $y = 4x - 2$, when $x = 0$, $y = -2$. $y = \frac{10}{2+3x^2} \Rightarrow x^2 = \frac{1}{3} \left(\frac{10}{y} - 2 \right) = \frac{2}{3} \left(\frac{5}{y} - 1 \right)$. Method 1: Use formula for volume of cone Required volume $= \frac{1}{3} \pi (1)^2 [2 - (-2)] + \pi \int_2^5 \frac{2}{3} \left(\frac{5}{y} - 1 \right) dy$ $= \frac{4}{3} \pi + \frac{2}{3} \pi [5 \ln y - y]_2^5$ $= \frac{2}{3} \pi [2 + (5 \ln 5 - 5) - (5 \ln 2 - 2)]$ $= \frac{2}{3} \pi \left(5 \ln \frac{5}{2} - 1 \right) \text{ units}^3$. Method 2: Use integration for volume of cone $y = 4x - 2 \Rightarrow x^2 = \left(\frac{y+2}{4} \right)^2$ Volume of cone $= \pi \int_{-2}^2 \left(\frac{y+2}{4} \right)^2 dy = \frac{\pi}{16} \left[\frac{(y+2)^3}{3} \right]_{-2}^2 = \frac{\pi}{48} (4^3) = \frac{4}{3} \pi \text{ units}^3$.
5 (i)	Method 1: $y = x^2$ ↓ Replace x by $x - b$ $y = (x - b)^2$ ↓ Replace x by ax $y = (ax - b)^2$ ↓ Replace y by $y - c$ $y - c = (ax - b)^2 \Rightarrow y = (ax - b)^2 + c$ The sequence of transformations are 1. translation in the positive x-direction by b units 2. scaling along the x-axis by a scale factor of $\frac{1}{a}$ 3. translation in the positive y-direction by c units Method 2: $y = x^2$

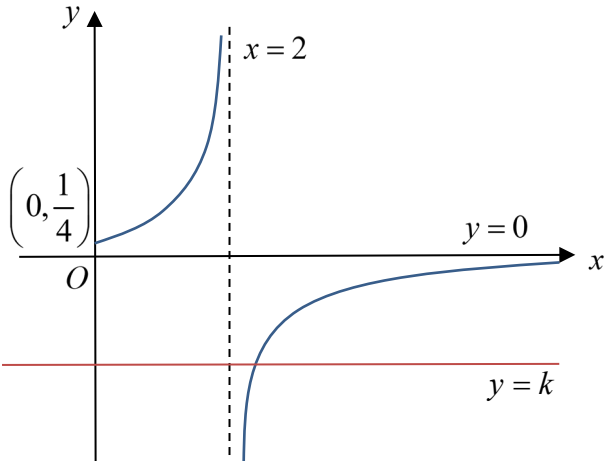
Qn	Solution
	↓ Replace x by ax $y = (ax)^2$ ↓ Replace x by $x - \frac{b}{a}$ $y = \left[a \left(x - \frac{b}{a} \right) \right]^2 \Rightarrow y = (ax - b)^2$ ↓ Replace y by $y - c$ $y - c = (ax - b)^2 \Rightarrow y = (ax - b)^2 + c$ The sequence of transformations are 1. scaling along the x-axis by a scale factor of $\frac{1}{a}$ 2. translation in the positive x-direction by $\frac{b}{a}$ units translation in the positive y-direction by c units
5 (ii)	<u>Method 1:</u> (1,1) ↓ Replace x by $x - b$ (1+b,1) ↓ Replace x by ax $\left(\frac{1+b}{a}, 1 \right)$ ↓ Replace y by $y - c$ $\left(\frac{1+b}{a}, 1+c \right)$ <u>Method 2:</u> (1,1) ↓ Replace x by ax $\left(\frac{1}{a}, 1 \right)$ ↓ Replace x by $x - \frac{b}{a}$ $\left(\frac{1}{a} + \frac{b}{a}, 1 \right) \Rightarrow \left(\frac{1+b}{a}, 1 \right)$ ↓ Replace y by $y - c$

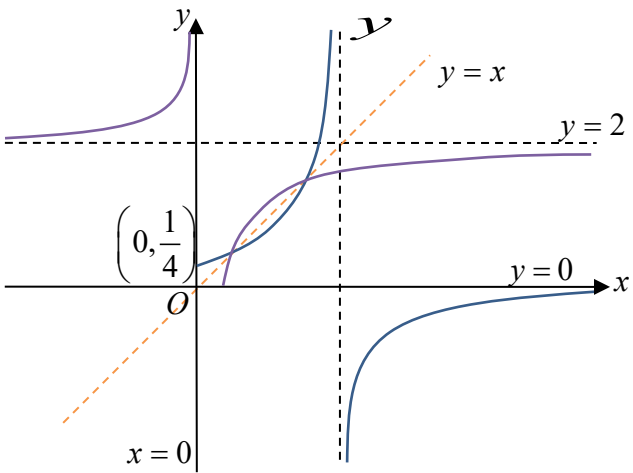
Qn	Solution
	$\left(\frac{1+b}{a}, 1+c\right)$ Comparing with (p, q), $p = \frac{1+b}{a}$ and $q = 1+c$. At $(0, r)$: $r = (0-b)^2 + c \Rightarrow r = b^2 + c \therefore p = \frac{1+b}{a}, q = 1+c, r = b^2 + c$
5 (iii)	$C_2 : y = (ax - b)^2 + c$ At $(2, 12)$: $(2a - b)^2 + c = 12 \dots (1)$ At $\left(\frac{1}{2}, 3\right)$: $\left(\frac{a}{2} - b\right)^2 + c = 3 \dots (2)$ $\frac{dy}{dx} = 2(ax - b)^{2-1}(a) = 2a(ax - b)$ At $\left(\frac{1}{2}, 3\right)$: $2a\left(\frac{a}{2} - b\right) = 0$ $a = 0$ (rejected since $a > 0$) or $a = 2b \dots (3)$ Method 1: Substituting (3) into (2): $\left(\frac{2b}{2} - b\right)^2 + c = 3 \Rightarrow c = 3$ Substituting (3) and $c = 3$ into (1): $(2 \times 2b - b)^2 + 3 = 12$ $\Rightarrow b^2 = 1 \Rightarrow b = -1$ (rejected since $b > 0$) or $b = 1$ Substituting $b = 1$ into (3): $a = 2(1) = 2$. Method 2: (1) - (2): $(2a - b)^2 - \left(\frac{a}{2} - b\right)^2 = 9$ Substituting (3) into the equation: $(2 \times 2b - b)^2 - \left(\frac{2b}{2} - b\right)^2 = 9$ $\Rightarrow b^2 = 1 \Rightarrow b = -1$ (rejected since $b > 0$) or $b = 1$ Substituting $b = 1$ into (3): $a = 2(1) = 2$. From (2): $\left(\frac{2}{2} - 1\right)^2 + c = 3 \Rightarrow c = 3$. Hence, $a = 2, b = 1, c = 3$.
6 (i)	$y = \sqrt{e^x} \cos x$ Method 1: Implicit differentiation $y^2 = e^x \cos x$ $2y \frac{dy}{dx} = (e^x)(-\sin x) + (\cos x)(e^x)$

Qn	Solution
	$2y \frac{dy}{dx} = y^2 - e^x \sin x. \text{ (shown)}$ Method 2: Direct differentiation $y = (e^x \cos x)^{\frac{1}{2}}$ $\frac{dy}{dx} = \frac{1}{2} (e^x \cos x)^{-\frac{1}{2}} [(e^x)(-\sin x) + (\cos x)(e^x)]$ $\frac{dy}{dx} = \frac{1}{2} \left(\frac{1}{y} \right) (y^2 - e^x \sin x)$ $2y \frac{dy}{dx} = y^2 - e^x \sin x. \text{ (shown)}$
6 (ii)	$2y \frac{dy}{dx} = y^2 - e^x \sin x \quad \dots \quad (1)$ $2y \frac{d^2 y}{dx^2} + \frac{dy}{dx} \left(2 \frac{dy}{dx} \right) = 2y \frac{dy}{dx} - [(e^x)(\cos x) + (\sin x)(e^x)]$ $2y \frac{d^2 y}{dx^2} + 2 \left(\frac{dy}{dx} \right)^2 = 2y \frac{dy}{dx} - (y^2 + e^x \sin x) \quad \dots \quad (2)$ When $x = 0$, $y = \sqrt{e^0 \cos 0} = 1$. From (1): $2(1) \frac{dy}{dx} = (1)^2 - e^0 \sin 0 \Rightarrow \frac{dy}{dx} = \frac{1}{2}$. From (2): $2(1) \frac{d^2 y}{dx^2} + 2 \left(\frac{1}{2} \right)^2 = 2(1) \left(\frac{1}{2} \right) - (1^2 + e^0 \sin 0)$ $\Rightarrow 2 \frac{d^2 y}{dx^2} + \frac{1}{2} = 0 \Rightarrow \frac{d^2 y}{dx^2} = -\frac{1}{4}.$ Hence, $y = 1 + \frac{\left(\frac{1}{2}\right)}{1!} x^1 + \frac{\left(-\frac{1}{4}\right)}{2!} x^2 + \dots$, i.e. $y = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots$
6 (iii)	Method 1: $y = \sqrt{e^x \cos x} = e^{\frac{1}{2}x} (\cos x)^{\frac{1}{2}}$ $= \left[1 + \left(\frac{1}{2}x \right) + \frac{1}{2!} \left(\frac{1}{2}x \right)^2 + \dots \right] \left(1 - \frac{x^2}{2} + \dots \right)^{\frac{1}{2}}$ $= \left(1 + \frac{1}{2}x + \frac{1}{8}x^2 + \dots \right) \left(1 - \frac{x^2}{4} + \dots \right)$ $= 1 - \frac{x^2}{4} + \frac{1}{2}x + \frac{1}{8}x^2 + \dots = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots \text{ (verified)}$

Qn	Solution
	Method 2: $y = \sqrt{e^x \cos x} = (e^x \cos x)^{\frac{1}{2}}$ $= \left[\left(1 + x + \frac{1}{2!}x^2 + \dots \right) \left(1 - \frac{x^2}{2} + \dots \right) \right]^{\frac{1}{2}}$ $= \left(1 - \frac{x^2}{2} + x + \frac{1}{2}x^2 + \dots \right)^{\frac{1}{2}}$ $= (1 + x + \dots)^{\frac{1}{2}}$ $= 1 + \frac{1}{2}x + \frac{\frac{1}{2}\left(-\frac{1}{2}\right)}{2!}x^2 + \dots = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \dots \text{ (verified)}$
7 (i)	Method 1: Use Cartesian equation $\Pi_1: 2x + y - 4z = 8$ Since $2(5) + (2) - 4(1) = 8$, A lies on Π_1. (verified) Method 2: Use scalar product form $\Pi_1: \mathbf{r} \cdot \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} = 8.$ Since $\overrightarrow{OA} \cdot \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} = 10 + 2 - 4 = 8$, A lies on Π_1. (verified)
7 (ii)	Let F be the foot of perpendicular from B to Π_1. Line $BF: \mathbf{r} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix}, \lambda \in \mathbb{R}.$ F on BF and $\Pi_1 \Rightarrow \begin{pmatrix} 1+2\lambda \\ \lambda-2 \\ 4-4\lambda \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} = 8$ $\Rightarrow 2 + 4\lambda + \lambda - 2 - 16 + 16\lambda = 8$ $\Rightarrow 21\lambda = 24 \Rightarrow \lambda = \frac{24}{21} = \frac{8}{7}.$ Hence, $\overrightarrow{OF} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} + \frac{8}{7} \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 7+16 \\ -14+8 \\ 28-32 \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 23 \\ -6 \\ -4 \end{pmatrix}.$ Required coordinates is $F \left(\frac{23}{7}, -\frac{6}{7}, -\frac{4}{7} \right).$

Qn	Solution
7 (iii)	Method 1: Hence method $\begin{aligned} \text{Required length} = \overrightarrow{BF} &= \left \frac{1}{7} \begin{pmatrix} 23 \\ -6 \\ -4 \end{pmatrix} - \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} \right \\ &= \frac{1}{7} \left \begin{pmatrix} 23-7 \\ -6+14 \\ -4-28 \end{pmatrix} \right = \frac{1}{7} \left \begin{pmatrix} 16 \\ 8 \\ -32 \end{pmatrix} \right \\ &= \frac{8}{7} \left \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} \right = \frac{8}{7} \sqrt{2^2 + 1^2 + (-4)^2} \\ &= \frac{8}{7} \sqrt{21} \text{ units.} \end{aligned}$ Method 2: Otherwise method $\overrightarrow{AB} = \begin{pmatrix} 1 \\ -2 \\ 4 \end{pmatrix} - \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ -4 \\ 3 \end{pmatrix}.$ $\begin{aligned} \text{Required length} &= \left \overrightarrow{AB} \cdot \frac{1}{\sqrt{2^2 + 1^2 + (-4)^2}} \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} \right \\ &= \frac{1}{\sqrt{21}} \left \begin{pmatrix} -4 \\ -4 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 1 \\ -4 \end{pmatrix} \right \\ &= \frac{1}{\sqrt{21}} -8 - 4 - 12 \\ &= \frac{24}{\sqrt{21}} = \frac{24}{21} \sqrt{21} = \frac{8}{7} \sqrt{21} \text{ units.} \end{aligned}$
7 (iv)	Method 1: Required angle $\begin{aligned} &= \sin^{-1} \left \frac{1}{\sqrt{1^2 + (-3)^2 + 1^2}} \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{(-1)^2 + 1^2 + 3^2}} \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right \\ &= \sin^{-1} \left \frac{1}{\sqrt{11}} \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} \cdot \frac{1}{\sqrt{11}} \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right \\ &= \sin^{-1} \left \frac{-1 - 3 + 3}{11} \right = \sin^{-1} \frac{1}{11} \approx 5.216^\circ = 5.2^\circ \text{ (1 d.p.)} \end{aligned}$

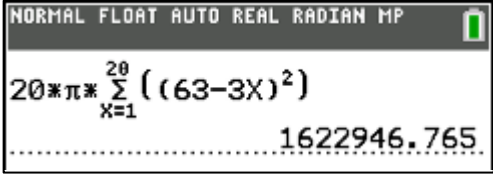
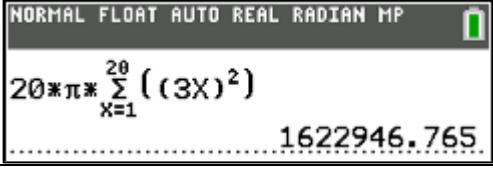
Qn	Solution
	Method 2: Required angle $= 90^\circ - \cos^{-1} \left \frac{1}{\sqrt{1^2 + (-3)^2 + 1^2}} \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} \bullet \frac{1}{\sqrt{(-1)^2 + 1^2 + 3^2}} \begin{pmatrix} -1 \\ 1 \\ 3 \end{pmatrix} \right $ $= 90^\circ - \cos^{-1} \frac{1}{11} \approx 5.216^\circ = 5.2^\circ \text{ (1 d.p.)}$
7 (v)	$\Pi_1 : 2x + y - 4z = 8 \quad \dots (1)$ $\Pi_2 : x - 3y + z = 2 \quad \dots (2)$ $\Pi_3 : -x + y + 3z = 5 \quad \dots (3)$ From graphing calculator, $x = \frac{149}{16}, y = \frac{29}{8}, z = \frac{57}{16}$. Required position vector is $\frac{149}{16}\mathbf{i} + \frac{29}{8}\mathbf{j} + \frac{57}{16}\mathbf{k}$.
8 (i)	 Since any horizontal line $y = k$, where $k \in \mathbb{R}$, $k < 0$ or $k \geq \frac{1}{4}$, intersects the graph of $y = f(x)$ at only one point, f is a one-one function. Hence, f^{-1} exists.
8 (ii)	Let $y = f(x) = \frac{1}{4 - x^2}$. $\Rightarrow 4 - x^2 = \frac{1}{y} \Rightarrow x^2 = 4 - \frac{1}{y}$ $\Rightarrow x = \sqrt{4 - \frac{1}{y}} \quad \text{or} \quad -\sqrt{4 - \frac{1}{y}} \quad (\text{rejected since } x \geq 0)$ $\Rightarrow f^{-1}(y) = \sqrt{4 - \frac{1}{y}} \Rightarrow f^{-1}(x) = \sqrt{4 - \frac{1}{x}}$ Domain of f^{-1} = Range of $f = (-\infty, 0) \cup \left[\frac{1}{4}, \infty\right)$ OR $\left\{x \in \mathbb{R} : x < 0 \text{ or } x \geq \frac{1}{4}\right\}$.

Qn	Solution
8 (iii)	 The graph of $y = f(x)$ is the reflection of the graph of $y = f^{-1}(x)$ in the line $y = x$. Method 1: $f(x) = f^{-1}(x) \Rightarrow f(x) = x$ $\frac{1}{4-x^2} = x \Rightarrow x(4-x^2) = 1$ $\therefore x^3 - 4x + 1 = 0. \text{ (shown)}$ Method 2: $f(x) = f^{-1}(x) \Rightarrow f^{-1}(x) = x.$ $\sqrt{4 - \frac{1}{x}} = x \Rightarrow \frac{4x-1}{x} = x^2$ $\therefore x^3 - 4x + 1 = 0. \text{ (shown)}$ From graphing calculator, $x = -2.11$ (rejected since $x \geq \frac{1}{4}$), 0.254, 1.86 (3 s.f.)
9 (i)	$AF = x \Rightarrow EF = CF = 1 - x$ By Pythagoras' Theorem, $\left(\frac{1}{2}\right)^2 + x^2 = (1-x)^2$ $\frac{1}{4} + x^2 = 1 - 2x + x^2 \Rightarrow x = \frac{3}{8}.$ Hence, $\overrightarrow{EF} = \overrightarrow{EA} + \overrightarrow{AF} = \frac{1}{2}\mathbf{a} + \frac{3}{8}\mathbf{b}$. (shown)
9 (ii)	$\overrightarrow{EF} \cdot \overrightarrow{EH} = \left(\frac{1}{2}\mathbf{a} + \frac{3}{8}\mathbf{b}\right) \cdot \left(-\frac{1}{2}\mathbf{a} + \mu\mathbf{b}\right) = 0 \text{ (since } \overrightarrow{EF} \perp \overrightarrow{EH})$ $\Rightarrow -\frac{1}{4}(\mathbf{a} \cdot \mathbf{a}) + \frac{1}{2}\mu(\mathbf{a} \cdot \mathbf{b}) - \frac{3}{16}(\mathbf{b} \cdot \mathbf{a}) + \frac{3}{8}\mu(\mathbf{b} \cdot \mathbf{b}) = 0$ $\Rightarrow -\frac{1}{4}(1) + \frac{3}{8}\mu(1) = 0 \text{ (since } \mathbf{a} \cdot \mathbf{a} = \mathbf{a} ^2 = \mathbf{b} ^2 = \mathbf{b} \cdot \mathbf{b} = 1 \text{ and } \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a} = 0)$ $\Rightarrow \frac{3}{2}\mu = \frac{1}{4} \Rightarrow \mu = \frac{2}{3}.$

Qn	Solution
9 (iii)	Line BC : $\mathbf{r} = \mathbf{b} + \alpha \mathbf{a}, \alpha \in \mathbb{R}$ Line EH : $\mathbf{r} = \frac{1}{2} \mathbf{a} + \beta \left(-\frac{1}{2} \mathbf{a} + \frac{2}{3} \mathbf{b} \right), \beta \in \mathbb{R}$
9 (iv)	Since P is on both lines, $\mathbf{b} + \alpha \mathbf{a} = \frac{1}{2} \mathbf{a} + \beta \left(-\frac{1}{2} \mathbf{a} + \frac{2}{3} \mathbf{b} \right)$ has a unique solution for α and β. $\Rightarrow \left(\frac{1}{2} - \alpha - \frac{1}{2} \beta \right) \mathbf{a} + \left(\frac{2}{3} \beta - 1 \right) \mathbf{b} = \mathbf{0}$ $\Rightarrow \begin{cases} \frac{1}{2} - \alpha - \frac{1}{2} \beta = 0 \\ \frac{2}{3} \beta - 1 = 0 \end{cases} \Rightarrow \begin{cases} \frac{1}{2} - \alpha - \frac{1}{2} \left(\frac{3}{2} \right) = 0 \\ \beta = \frac{3}{2} \end{cases} \Rightarrow \begin{cases} \alpha = -\frac{1}{4} \\ \beta = \frac{3}{2} \end{cases}$ Hence, $\overrightarrow{OP} = \mathbf{b} - \frac{1}{4} \mathbf{a}$.
10 (i)	$\frac{d^2x}{dt^2} = 9 - 5t \Rightarrow \frac{dx}{dt} = 9t - \frac{5}{2}t^2 + b$ $\Rightarrow x = \frac{9}{2}t^2 - \frac{5}{6}t^3 + bt + c, \text{ where } b \text{ and } c \text{ are arbitrary constant.}$
10 (ii)	When $t = 0, x = 1$, $x = \frac{9}{2}t^2 - \frac{5}{6}t^3 - \frac{9}{2}t + c \Rightarrow c = 1.$ When $t = 3, \frac{dx}{dt} = 0$, $\frac{dx}{dt} = 9t - \frac{5}{2}t^2 + b \Rightarrow b = -\frac{9}{2}.$ Hence, $x = \frac{9}{2}t^2 - \frac{5}{6}t^3 - \frac{9}{2}t + 1$.
10 (iii)	$\frac{dx}{dt} = \frac{k}{x^2}$
10 (iv)	$\frac{dx}{dt} = \frac{k}{x^2}$ $\Rightarrow x^2 \frac{dx}{dt} = k$ $\Rightarrow \int x^2 dx = \int k dt$ $\Rightarrow \frac{x^3}{3} = kt + c$ When $t = 0, x = 1, c = \frac{1}{3}$ When $t = 1, x = 4$, $\frac{4^3}{3} = k + \frac{1}{3} \Rightarrow k = 21$

Qn	Solution
	Hence, $\frac{x^3}{3} = 21t + \frac{1}{3}$ $\Rightarrow x^3 = 63t + 1$. $\Rightarrow t = \frac{1}{63}(x^3 - 1)$ When $x = 16$, $t = \frac{1}{63}(16^3 - 1) = 65$. It takes 65 months for the insect population to reach 16000.
11(i)	Let the radius of the nth layer be T_n $T_n = 60 + (n-1)(-3) > 0$ $60 - 3n + 3 > 0$ $3n < 63$ $n < 21$ The maximum number of layers is 20.
11(ii)	$T_{20} = 60 + (20-1)(-3) = 3$ <u>Method 1(a)</u> The total length of ribbon required $= 2\pi(60) + 2\pi(57) + 2\pi(54) + \dots + 2\pi(3)$ $= 2\pi(60 + 57 + 54 + \dots + 3)$ $= 2\pi \times \frac{20}{2} [2(60) + (20-1)(-3)]$ $= 1260\pi \approx 3958.406 = 3960 \text{ cm (3 s.f.)}$ <u>Method 1(b)</u> The total length of ribbon required $= 2\pi(60) + 2\pi(57) + 2\pi(54) + \dots + 2\pi(3)$ $= 2\pi(3) + 2\pi(6) + 2\pi(9) + \dots + 2\pi(60)$ $= 2\pi(3 + 6 + 9 + \dots + 60)$ $= 2\pi \times \frac{20}{2} [2(3) + (20-1)(3)]$ $= 1260\pi \approx 3958.406 = 3960 \text{ cm (3 s.f.)}$ <u>Method 2(a)</u> The total length of ribbon required $= \pi(120) + \pi(114) + \pi(108) + \dots + \pi(6)$ $= \pi(120 + 114 + 108 + \dots + 6)$ $= \pi \times \frac{20}{2} [2(120) + (20-1)(-6)]$ $= 1260\pi \approx 3958.406 = 3960 \text{ cm (3 s.f.)}$

Qn	Solution
	<u>Method 2(b)</u> The total length of ribbon required $= \pi(120) + \pi(114) + \pi(108) + \dots + \pi(6)$ $= \pi(6 + 12 + 18 + \dots + 120)$ $= \pi \times \frac{20}{2} [2(6) + (20-1)(6)]$ $= 1260\pi \approx 3958.406 = 3960 \text{ cm (3 s.f.)}$ <u>Method 3</u> Radius of the nth layer $= 60 + (n-1)(-3) = 63 - 3n$ Circumference of the nth layer $= 2\pi(63 - 3n)$ The total length of ribbon required $= \sum_{n=1}^{20} 2\pi(63 - 3n)$ $\approx 3958.406 = 3960 \text{ cm (3 s.f.)}$
11(iii)	<u>Method 1</u> Radius of the nth layer $= 60 + (n-1)(-3) = 63 - 3n$. Volume of the nth layer $= \pi(63 - 3n)^2(20)$ The total volume of the structure $= \sum_{n=1}^{20} \pi(63 - 3n)^2(20)$ $= 20\pi \sum_{n=1}^{20} (63 - 3n)^2, \text{ where } p = 20 \text{ and } f(n) = 63 - 3n$ OR $= 180\pi \sum_{n=1}^{20} (21 - n)^2, \text{ where } p = 180 \text{ and } f(n) = 21 - n$ <u>Method 2</u> The volume of the uppermost layer $= \pi(3)^2(20)$ The volume of the 2nd uppermost layer $= \pi(6)^2(20)$ The volume of the 3rd uppermost layer $= \pi(9)^2(20)$ The total volume of the structure $= \pi(3)^2(20) + \pi(6)^2(20) + \pi(9)^2(20) + \dots + \pi(60)^2(20)$ $= 20\pi(3^2 + 6^2 + 9^2 + \dots + 60^2)$ $= 20\pi \sum_{n=1}^{20} (3n)^2, \text{ where } p = 20 \text{ and } f(n) = 3n$ OR $= 180\pi \sum_{n=1}^{20} n^2, \text{ where } p = 180 \text{ and } f(n) = n$

Qn	Solution								
	Last part----- Using GC, (Method 1) The total volume of the structure = $20\pi \sum_{n=1}^{20} (63 - 3n)^2 = 1622946.765$. Since $1622946.765 > 1\,500\,000$, Arthur does not have enough amount of clay to construct a 20-layered structure.  Using GC, (Method 2) The total volume of the structure = $20\pi \sum_{n=1}^{20} (3n)^2 = 1622946.765$. Since $1622946.765 > 1\,500\,000$, Arthur does not have enough amount of clay to construct a 20-layered structure. 								
11(iv)	The height of the first layer = 20 and the common ratio is 0.8. The total height of the structure ≤ 85 $\frac{20(1 - 0.8^n)}{1 - 0.8} \leq 85$ Method 1: Algebraic Method $1 - 0.8^n \leq 0.85$ $0.8^n \geq 0.15$ $n \ln 0.8 \geq \ln 0.15$ $n \leq \frac{\ln 0.15}{\ln 0.8} \quad (\text{Note: } \ln 0.8 < 0)$ $n \leq 8.51$ The maximum number of layers is 8. Method 2: GC Table <table border="1" data-bbox="328 1697 707 1906"> <thead> <tr> <th>n</th><th>$\frac{20(1 - 0.8^n)}{1 - 0.8}$</th></tr> </thead> <tbody> <tr> <td>7</td><td>79.028</td></tr> <tr> <td>8</td><td>83.223</td></tr> <tr> <td>9</td><td>86.578</td></tr> </tbody> </table> Maximum number of layers is 8.	n	$\frac{20(1 - 0.8^n)}{1 - 0.8}$	7	79.028	8	83.223	9	86.578
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7	79.028								
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